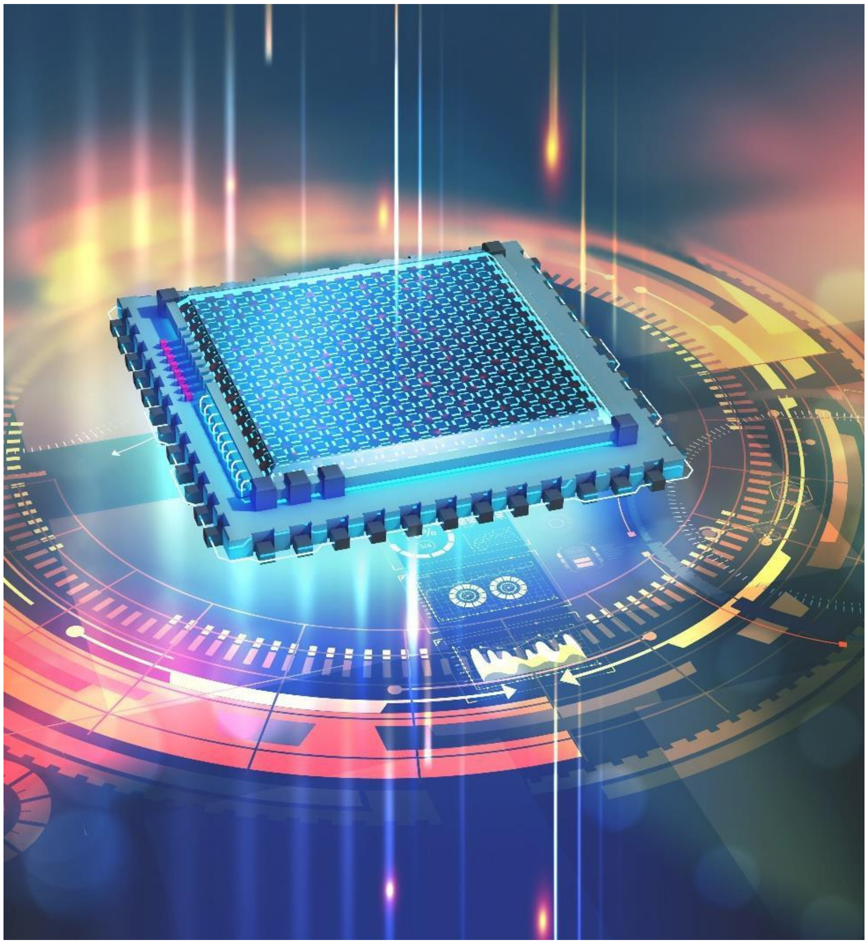
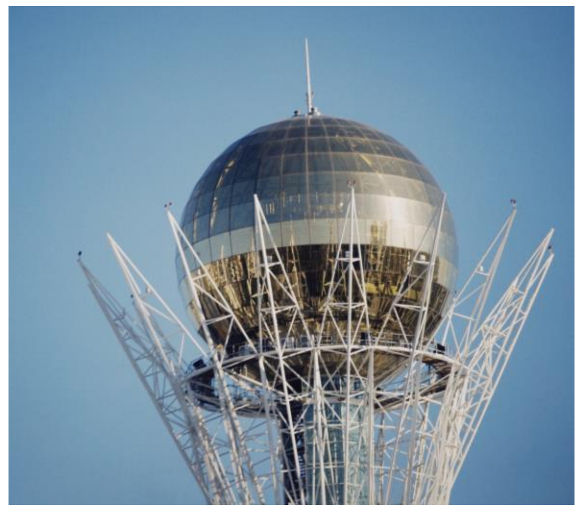
Model RFS-4220T
RF & Microwave Signal Generator





Features

- Wide Frequency Range (100MHz-22GHz)
- Adjustment Level up to +15 dBm
- Low Phase Noise & Excellent Spectral Purity
- Precision Frequency Control
- Desktop & LAN control

Model RFS-4220T

100 MHz – 22 GHz

Applications

- Automated Test Environment
- Antenna Design
- Aerospace & Defense Research
- Wireless Infrastructure Design
- Satellite Link Testing

INTRODUCTION

The RFS-4220T is a compact, high-performance RF and microwave signal generator designed for lab testing, automated systems, and RF development. Offering precise frequency control, low phase noise, and a broad operating range, this generator provides a cost-effective and user-friendly solution for engineers and researchers.

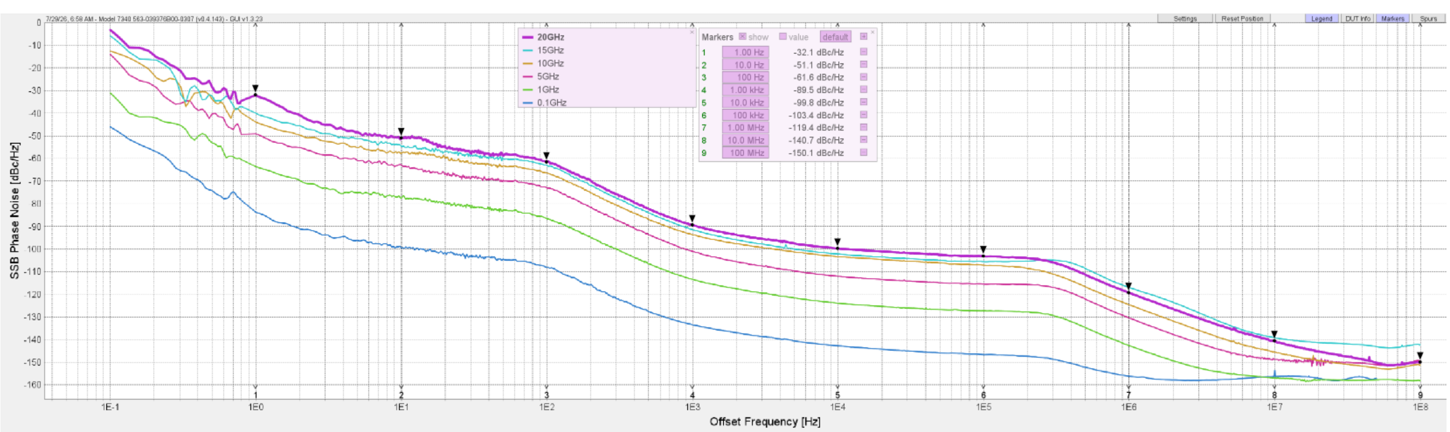
Key Features

- Wide frequency coverage: 0.1 GHz to 20 GHz calibrated, settable to 22 GHz
- Flexible and user-friendly design: small, portable aluminum enclosure ideal for lab and field use
- High output power: up to +15 dBm with fine power control
- Low phase noise: excellent spectral purity for demanding applications
- Precision frequency control: ultra-fine tuning with small frequency step sizes (1 Hz)
- Dual control options: front panel touchscreen and buttons, or remote LAN control

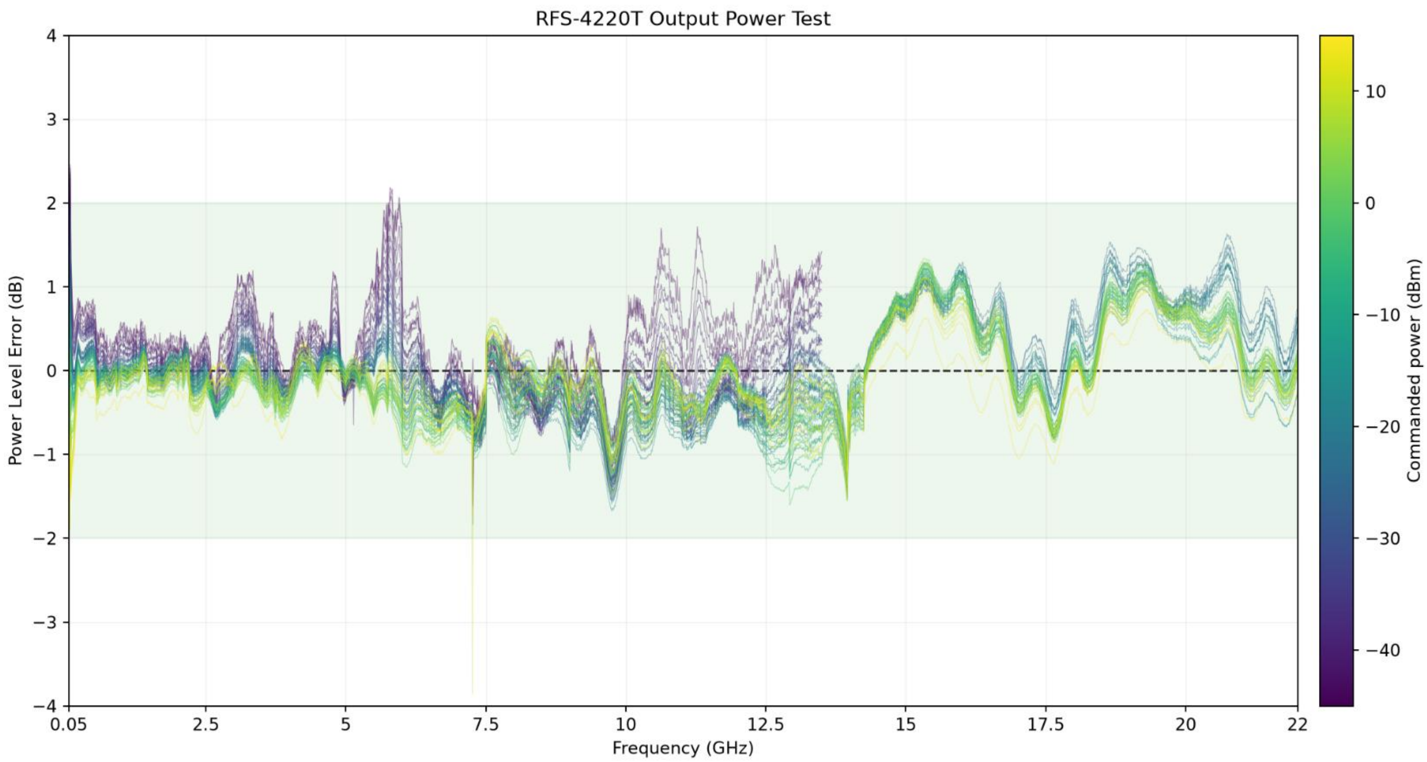
PRODUCT SPECS

Feature	**RFS-4220T**
Frequency Range	0.1 GHz – 20 GHz calibrated, settable up to 22 GHz
Power Range	-45 to +15 dBm (0-13.5 GHz) -26 to +15 dBm (13.5-22 GHz)
Power Accuracy	±2.0 dB typical
Frequency Step Size	1 Hz
Harmonic Content	< -25 dBc (typical)
Reference Source	Internal 10.0MHz OCXO or external 10.0 MHz reference
Internal Reference Stability	±10.0 ppb
Control Interface	Touchscreen, front panel buttons, or computer application
Power Supply	90–264 VAC, 50/60 Hz, 100 VA max
Size & Enclosure	8.6 × 4 × 12.25 in (21.84 × 10.16 × 31.12 cm)
Weight	5.6 lbs (2.55 kg)

PHASE NOISE



POWER ACCURACY



APPLICATIONS

Bench and laboratory

General RF Lab Use

One instrument covers the bench from 0.1 GHz through 20 GHz calibrated and sets as high as 22 GHz, at levels from -45 dBm up to +15 dBm. The touchscreen, the front-panel buttons and the rotary knob each drive the full instrument, so a quick check never waits on a PC to boot. At 5.6 lbs and 8.6 by 4 by 12.25 inches it moves between benches without a cart.

Educational / University Lab Use

A student can produce a calibrated signal in the first minute of a lab session, because nothing has to be installed first. The same unit then teaches remote control properly, since every front-panel function is reachable over SCPI when the course moves on to automation. One box spans VHF through microwave, so a teaching lab does not need three.

Flexible LO Sourcing

Low phase noise and an internal 10 MHz OCXO holding 10 ppb make this a clean local oscillator for mixers, converters and receiver front ends. Feed the external 10 MHz input from the house standard when the measurement has to stay coherent with everything else on the bench. Fine level control puts drive exactly where the mixer wants it.

Up-converting and Down-converting

As the LO behind a mixer, this generator sets the conversion. A 1 Hz step size lets you land an intermediate frequency precisely rather than trimming afterwards, and the OCXO holds it there through the measurement. Harmonics are specified at -25 dBc typical, so plan a filter when a conversion demands a purer tone.

Automation and production

Automated Testing Environments

SCPI over LAN or USB puts every function under program control, and the external trigger input lets the source step in time with the rest of the rack instead of being polled. Sweep lists live in internal storage, so a station recalls a saved CSV rather than rebuilding the sweep at run time. Multi-session control lets more than one host share the instrument without a switchbox.

APPLICATIONS

Production Verification and Test Setups

Automatic network discovery means a replacement unit joins the line without anyone editing an address into a test script. Saved sweep lists make the stimulus identical on Monday and on Friday, which is what makes a production result comparable. Power accuracy of 2.0 dB typical sets a known bound when you write the limits.

Antenna, EMC and compliance

Antenna Design

Run a stored sweep list while the antenna rotates, then recall the identical sweep for the next build so two patterns can honestly be compared. Coverage to 22 GHz reaches modern patch, horn and array work in one instrument. Output to +15 dBm supplies useful drive for chamber and near-field measurements.

EMC Testing

For pre-compliance immunity work, the generator steps through a saved frequency list while a monitor watches the product under test. The external trigger keeps each step aligned with dwell and measurement timing, so the log lines up with the stimulus. Knowing the harmonic content is -25 dBc typical tells you when to add a filter and keep a failure attributable to the product rather than the source.

Communications, radar and satellite

802.11n Development / Testing

This is a CW and swept source, so in a wireless lab its job is the interferer and the local oscillator rather than the modulated waveform. Place a clean tone in the 2.4 GHz or 5 GHz bands to measure receiver desensitization, adjacent-channel rejection and blocker performance. Levels down to -45 dBm let sensitivity work happen at realistic amplitudes.

5G Testing

The same role applies to sub-6 GHz and mid-band 5G development: a stable CW stimulus, a clean LO for a converter stage, or a controlled blocker for receiver characterization. Frequency coverage well past the FR1 bands leaves headroom for harmonic and spurious investigations that stop at the band edge on a narrower source.

Ku-band Satellite Link Testing

Coverage reaches both the roughly 12 GHz downlink and 14 GHz uplink ranges from a single box, which is the awkward span for many bench sources. Note that the output range changes above 13.5 GHz to -26 dBm through +15 dBm, so check the uplink side against your link budget. Lock to a station reference where the measurement has to stay coherent.

X-Band Radar Applications

The 8 GHz to 12 GHz band sits comfortably inside the calibrated range, with the phase noise performance a radar receiver chain expects from its local oscillator. Stored sweeps repeat the same frequency profile across builds and across shifts. The external trigger ties the source to the radar's own timing rather than to a stopwatch.

Line of Sight Link Testing

Point-to-point links need a source that holds a frequency and a level for as long as the test runs, then returns to exactly the same settings tomorrow. The OCXO reference and saved instrument states cover both. Coverage to 22 GHz reaches the licensed microwave bands these links occupy.

Wireless Infrastructure Design

During base station and repeater development the generator serves as the reference tone for gain and flatness, the LO for a conversion stage, or the unwanted signal for isolation and rejection tests. Remote control over LAN keeps it in the rack while the engineer works from a desk. One box covers the cellular bands and the microwave backhaul above them.

Transponder Verification

Interrogate a transponder with a precisely placed carrier, and step through a saved list to confirm the response across its operating range. The 1 Hz step size resolves narrow channel plans, and level control from -45 dBm establishes the threshold at which the unit still answers.

Aerospace / Defense Research

Programs in this space usually need reach, repeatability and a documented reference. Coverage to 22 GHz, an OCXO time base with an external reference input, and SCPI control provide all three from a benchtop chassis that runs on standard mains. Saved sweep lists keep a test reproducible long after the engineer who wrote it has moved on.